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St. Joseph's College of Arts & Science (Autonomous)
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MT102Q – TWO DIMENSIONAL & THREE DIMENSIONAL GEOMETRY

Time : 3 hrs **Max Marks :75**

SECTION – A (5X2=10)

Answer **ALL** Questions

1. In hyperbola the value of e is ……..
2. Write down Angle between the planes.
3. Define coplanar lines.
4. Find the radius of the Sphere
   $2x^2 + 2y^2 + 2z^2 + 6x - 6y + 8z + 9 = 0$.
5. Define a right circular cone.

SECTION – B (3X5=15)

Answer any **THREE** Questions

6. Find the equation of the Parabola whose focus is (1, -1) and the direction is a $x + y + 7 = 0$.
7. Find the equation of the Plane through the point (4, 4, 4) and through the line AB where A is (1, 1, 1) and the direction ratios of AB are (3, 2, 1).
8. Find the equation of the line of intersection of the planes
   $2x - 3y + z + 4 = 0$; $x + y + 2z - 3 = 0$ at the point (1, 2, 0).
9. Find the equation of the Sphere whose centre is (1, 1, 1) and which passes through (3, 3, 2).
10. Obtain the equation of the right circular cone having the line
    $x = y = z$ as its axis and the line $2x = y = -3z$ as a generator.

SECTION - C (5X10=50)

Answer **ALL** Questions

11. a) Find the locus of the poles of all tangents to the parabola $y^2 = 4ax$ with respect to the parabola $y^2 = 4bx$.

(or)

b) Prove that the locus of the poles of all normal chords of the rectangular Hyperbola $xy=c^2$ is the curve $(x^2-y^2)^2+4c^2xy=0$.

12. a) Find the equation of the plane through the line of intersection of the planes $x+2y+3z+2 = 0$ and $2x+3y-z+3 = 0$ and perpendicular to the plane $x+y+z = 0$.

(or)

b) Find the equation of the plane through the point (0,-5,1) and perpendicular to the Planes $x+4y+2z+3=0$; $2x+5y+3z+4=0$.

13. a) Find the shortest distance between the lines $3x-9y+5z = 0 = x+y-z$; $6x+8y+3z-10 = 0 = x+2y+z-3$

(or)

b) Show that the following lines are co planar and find the equation of the plane of co-planarity $\frac{x-7}{2} = \frac{y-10}{3} = \frac{z-13}{4}$; $\frac{x-3}{1} = \frac{y-5}{2} = \frac{z-7}{3}$.

14. a) Find the equation of the sphere through the circle $s=x^2+y^2+z^2+4x-4y+2z=0$; $s'=x^2+y^2+z^2-2x-2y+2z=0$ and having the cube on the plane $x+y+z+5=0$

(or)

b) Find the equation of the sphere through (0,0,0), (a,0,0),(0,b,0) (0,0,c).

15. a) Find the equation of the cone whose vertex is the point (1,1,0) and whose guiding curve is $x^2 + y^2 = 4 , y = 0$

(or)

b) Find the equation of the quadratic cone through the x, y, z axes and the lines through the origin with direction ratios 2,4,1 ;3,3,1.

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